Neuro-Symbolic Detection of Cyber Intimate Partner Violence: A Graph-RAG Approach Integrating Clinical Risk Assessment Questionnaires and Legal Grounding

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Abstract

The automatic detection of Cyber Intimate Partner Violence (CIPV) requires models capable of understanding subtle coercive patterns beyond explicit toxicity. Traditional classifiers often fail to capture the context-dependent nature of abuse, such as gaslighting, isolation, or economic control. This report presents a novel *Context-Aware Graph-RAG* pipeline that integrates Large Language Models (LLMs) with a domain-specific Knowledge Graph (KG) built on Neo4j. The system is grounded in a dual-layer taxonomy: a psychological set of abuse tactics derived from a high-fidelity synthetic dataset (Posa et al., 2026) [1] and a clinical assessment layer based on the **SARA (Spousal Assault Risk Assessment)** factors [2]. Furthermore, the model is enriched with legal definitions from the Italian "Codice Rosso" legislation [3]. A comparative benchmark on state-of-the-art models (Qwen3-32B [4], GPT-OSS-20B [5], Llama-3.3-70B-Versatile [6]) demonstrates that our neuro-symbolic approach significantly enhances explainability and risk assessment accuracy compared to standard black-box classifiers, effectively bridging the gap between linguistic cues and clinical evidence, while mitigating LLM hallucinations.

Keywords

Graph-RAG, Explainable AI, SARA Questionnaire, Cyber Intimate Partner Violence, Neo4j, Legal NLP, Neuro-Symbolic Integration

1. Introduction

Intimate Partner Violence (IPV) has increasingly moved into the digital sphere, manifesting as Cyber Intimate Partner Violence (CIPV). This form of abuse is particularly insidious because it often lacks explicit profanity, hate speech, or obvious "toxic" keywords. Instead, it relies on patterns of *coercive control*, where a partner uses digital surveillance, social isolation, and emotional manipulation to erode the victim's autonomy and psychological safety.

Current NLP models, despite their size, face two primary hurdles in this domain:

1. **Context sensitivity:** a message like "Where are you?" is neutral in isolation but becomes a marker of stalking or location control if repeated obsessively or sent during a sensitive relationship stage, such as a termination attempt.
2. **Explainability and legal grounding:** forensic analysts, police officers, and psychologists require systems that don't just provide a generic "toxicity score," but explain *why* a specific behavior is abusive according to established clinical criteria (like SARA) and which specific article of the law it violates. As highlighted by Hogan et al. [7], LLMs excel at language generation but act as opaque black boxes, whereas KGs offer structural transparency, correctness, and explicit reasoning paths. Furthermore, recent literature reviews [8] emphasize that pre-trained language models struggle with structured reasoning and lack interpretability. Our approach resolves these issues by delegating the legal and clinical reasoning to the explicit structure of the KG, forcing the LLM to generate transparent and traceable analyses.

Our project addresses these issues by proposing a **Context-Aware Graph-RAG** system. To prevent the model from hallucinating non-existent laws or clinical factors, a known vulnerability of LLMs when dealing with complex domains [9], we integrate a structured Knowledge Graph built on Neo4j.

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Containing nodes for abuse tactics, psychological archetypes, clinical risk factors, and legal articles, this KG is coupled with the reasoning power of SOTA LLMs (Llama-3.3-70B). We provide a model that effectively "consults the expert manuals" before classifying a chat. This ensures that every detection is not just a statistical guess, but a justified, traccable forensic assessment.

2. Related work

This project builds upon recent research aimed at overcoming the scarcity of private IPV data while ensuring clinical and legal rigor.

2.1. Neuro-Symbolic integration and hallucination mitigation

Integrating Knowledge Graphs (KGs) into LLMs is the current gold standard for reducing hallucinations. We adopt strategies like *Knowledge-Aware Inference* [9] and "Amalgamation" [7] to provide a deterministic forensic boundary (Codice Rosso and SARA) for the LLM. This follows the recommendations of AlKhamissi et al. [10], who argue against using LLMs as implicit knowledge bases due to consistency issues. By separating the inferential engine from the Neo4j KG, we ensure that the model operates within a verifiable normative framework.

2.2. Explainable AI (XAI) and the legal "black-box" problem

Automated decision-making in the legal domain requires high levels of accountability [11]. Traditional "black-box" deep learning models often lack the interpretability required by legal adjudicators. Our architecture replaces opaque parametric memory with the deterministic retrieval of "expert knowledge" (SARA factors [2] and Penal Code articles [3]). To align with legal standards, our model generates forensic reports with an explicit **Confidence Score**. Outputs with less than 50% semantic overlap trigger a mandatory warning for human inspection, ensuring the system acts as a decision-support tool rather than an autonomous adjudicator.

2.3. Hybrid Graph-RAG architectures

Recent studies demonstrate that combining semantic vector search with explicit graph traversal (GraphRAG) significantly outperforms standard RAG frameworks in knowledge-intensive tasks [12]. We transposed this hybrid approach—proven effective in complex domains like medicine—into the informal linguistic environment of cyber abuse. By utilizing a specialized high-dimensional embedding model (**BAAI/bge-m3**) for dense retrieval and Neo4j for sparse lexical matching, our system provides the LLM with highly accurate clinical and legal context.

2.4. Knowledge bases and media framing

We utilize the *IDaToC* taxonomy [13], which categorizes toxic conversations into 10 psychological archetypes (e.g., *Narcissist and Submissive*) to provide systemic relationship context. Furthermore, we integrate the framing analysis from *FMNews* [14], which analyzed Italian media reporting on gender-based violence, bridging the gap between public news discourse and private chat terminology.

3. Synthetic dataset generation pipeline

This project builds upon a robust foundation of recent research aimed at overcoming the scarcity of private and sensitive IPV data due to obvious ethical and privacy constraints. Specifically, our work utilizes the dataset generated by Posa et al. [1], who developed a sophisticated **Stage-Wise LLM Pipeline** to address the fact that real-world relational data is notoriously scarce, a problem widely acknowledged in KG extraction literature [15]. The dataset consists of 24 high-fidelity synthetic

conversations transformed from public domain news (FMNews) into realistic private chats through four rigorous steps:

1. **News processing:** real-world femicide cases were aggregated and rewritten into chronological macro-reports to serve as a factual and emotional anchor.
2. **Scenario construction:** an LLM (GPT-5.2) acted as a forensic analyst to extract structured *Scenario Cards*. This defined the relationship type, the specific **abuse tactics** involved (e.g., Jealousy, Location Control), and the **relationship stage** (e.g., Terminating).
3. **Persona creation:** to avoid generic dialogues, a "Clinical Psychologist" LLM (Grok-3) generated deep psychological profiles (*Persona Cards*) grounded in the **Big Five** personality traits and Attachment Theory.
4. **Chat generation:** to maximize conversational realism and explore the boundaries of LLM capabilities, Posa et al. implemented and compared three distinct generation strategies:
   - **Strategy 1 (Single-shot):** a single prompt generates the entire dialogue, resulting in a more didactic and structurally predictable interaction.
   - **Strategy 2 (Single-LLM roleplay):** a single instance alternates between victim and abuser roles turn-by-turn.
   - **Strategy 3 (Dual-LLM roleplay):** two separate, consistently assigned model instances simulate the dyad, resulting in chaotic, non-linear, and highly realistic interactions.

Our work acts as the downstream detection module for this pipeline, utilizing the defined taxonomies as the ontological backbone for our Knowledge Graph. By leveraging these three distinct strategies, we performed a **Generation Strategy Ablation** to analyze whether our Graph-RAG engine maintains superior precision and confidence across both didactic (Strategy 1) and realistic/chaotic (Strategy 3) conversational data.

4. Proposed approach

The pipeline integrates a structured Knowledge Graph, a context-aware preprocessing layer, and a multi-stage Graph-RAG inference engine.

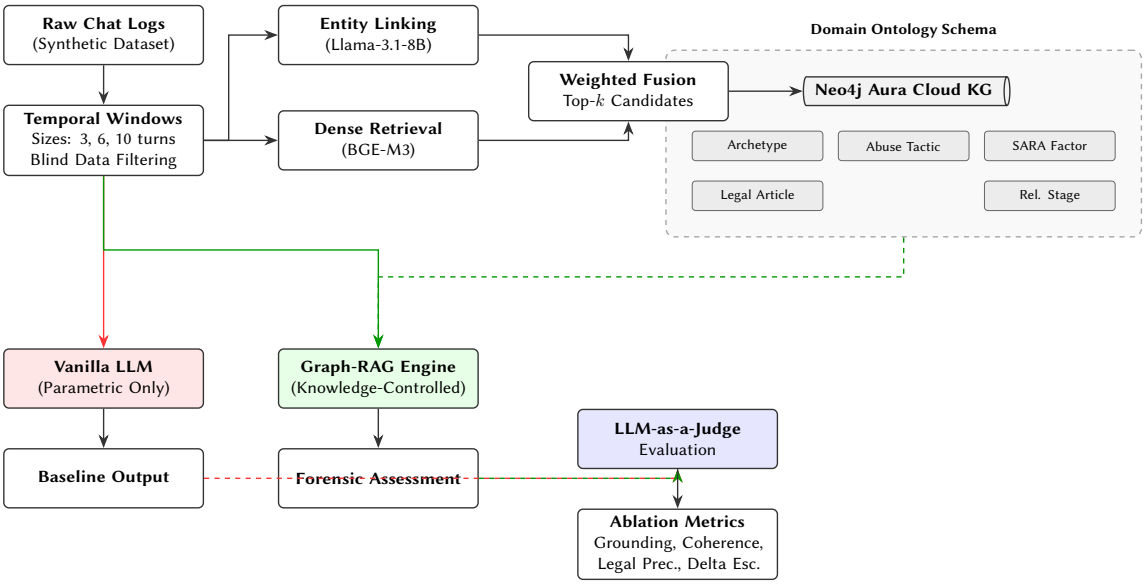

Figure 1: The Context-Aware Graph-RAG pipeline and ablation architecture. Raw chats are segmented into expanding temporal windows (Sizes: 3, 6, 10). The data follows two paths: **Path A (Green):** The RAG Engine constrains the LLM with expert context retrieved via Weighted Hybrid Search. **Path B (Red):** The raw chat is fed into a Vanilla LLM (Ablation Baseline). Both outputs are evaluated by the *LLM-as-a-Judge* module to compute comparative metrics.

4.1. Data preprocessing and multi-scale contextualization

Before analysis, the raw synthetic dataset undergoes a four-stage preprocessing pipeline to transform structured JSON data into a format suitable for forensic linguistic analysis while ensuring scientific integrity.

1. **Narrative flattening:** the original dataset is stored as structured JSON objects. To allow the LLM to perceive conversational flow, these objects are "flattened" into narrative strings in the format: `[Sender]: [Content]`. This allows for the distinction between the perpetrator's coercive patterns and the victim's reactive signals.
2. **Blind data filtering:** yo ensure the integrity of the detection task, a "Blind Testing" protocol is applied. All original metadata—such as `scenario_id`, `perpetrator_tactic`, and labels—are stripped. The system identifies violence solely through linguistic cues and KG grounding, effectively mimicking a real-world forensic investigation.
3. **Lexical normalization:** a regex-based normalization layer is applied to remove system-specific tags (e.g., <*think*> tags) and newline artifacts. This ensures that high-dimensional embeddings generated by the BGE-M3 model focus exclusively on semantic content.
4. **Multi-scale sliding window segmentation:** single-sentence analysis is insufficient for detecting the cumulative nature of coercive control. A sliding window mechanism is implemented for each turn t, constructing a query context by aggregating the current message with its history. Three distinct window sizes (N) are evaluated:
   - **Test A (Baseline):** `WINDOW_SIZE = 3`. Focuses on immediate "action-reaction" cycles (e.g., a threat following a boundary-setting attempt).
   - **Test B (Medium):** `WINDOW_SIZE = 6`. Designed to identify rapidly escalating conflict and early signs of gaslighting.
   - **Test C (Broad):** `WINDOW_SIZE = 10`. Allows the model to detect long-term patterns of isolation and cumulative emotional manipulation invisible in shorter sequences.

4.2. Knowledge graph construction and infrastructure

A specialized Knowledge Graph (KG) was developed and deployed on the **Neo4j Aura Cloud** infrastructure to encode the deterministic domain knowledge required for forensic analysis. The graph schema consists of 37 nodes categorized into five ontological domains:

- **Category:** 10 relational psychological archetypes derived from IDaToC (e.g., *Narcissist and Submissive*) [13].
- **Tactic:** 10 specific abuse mechanisms (e.g., *Gaslighting*, *AI Manipulation*).
- **SARA Factor:** 10 clinical indicators from the *Spousal Assault Risk Assessment* protocol (e.g., *S12: Extreme Terror*) [2].
- **Stage:** 4 relationship lifecycle phases (e.g., *Differentiation*, *Final Phase and Peril*).
- **Law:**mMapping of the Italian *Codice Rosso* framework, including articles 572, 612-bis, 612-ter, and 612-quater c.p [3].

Structural reasoning is performed via directed arcs such as `CHARACTERISTIC_OF` and `TRIGGERS_SARA`, enabling the engine to execute multi-hop traversals. For instance, detecting "AI Manipulation" leads deterministically to the relevant penal provision and associated clinical risk factors.

4.3. The Graph-RAG engine

The system builds upon the Graph-RAG architecture formalized by Edge et al. [12], allowing the LLM to reason over interconnected entities rather than localized facts.

4.3.1. Weighted hybrid retrieval strategy

The retrieval engine utilizes a **Weighted hybrid search** strategy, which ensures technical precision in legal contexts through three components:

1. **Dense retrieval (semantic search):** the **BAAI/bge-m3** model is utilized to generate 1024-dimensional embeddings. BGE-M3 captures fine-grained nuances between Italian legal terminology (e.g., *Atti Persecutori*) and colloquial language.
2. **Sparse Retrieval (Entity linking):** a lightweight LLM (Llama-3.1-8B) dynamically extracts key forensic entities (e.g., "GPS," "threat"). These are matched against KG nodes using the Cypher CONTAINS operator.
3. **Hybrid scoring fusion:** candidates are ranked by a **Unified Weighted Score** based on semantic cosine similarity ($vScore$) and entity overlap ($kScore$). This ensures retrieved nodes are both semantically relevant and lexically anchored to the specific chat.

4.4. Multi-model benchmark

To identify the most reliable inferential engine for forensic analysis, a comparative benchmark was conducted between three state-of-the-art models: **Llama-3.3-70B-Versatile** [6], **Qwen3-32B** (R1-Distill) [4], and **GPT-OSS-20B** [5]. The models were evaluated on critical chat windows to establish the performance ceiling for forensic reasoning in CIPV. These specific versions were selected to cover three distinct architectural paradigms: high-capacity general reasoning (Llama), explicit reasoning-based "Chain-of-Thought" (Qwen), and structured forensic synthesis (GPT-OSS), allowing for a comprehensive evaluation of different LLM "philosophies" against Knowledge Graph constraints.

Qwen3-32B [4] exhibited impressive logic-deductive capabilities through its internal "Chain-of-Thought" (CoT) processing, evidenced by its *<think>* tags. However, it was discarded due to a tendency toward *legal hallucination*. While its reasoning was sound, it frequently confused specific Italian articles (e.g., misclassifying Art. 612-ter as Stalking, which is legally defined under Art. 612-bis) [3]. Furthermore, its Italian linguistic register was found to be too schematic and less formal than required for a professional psychological report.

GPT-OSS-20B [5] demonstrated a strong capability for structured formatting, automatically generating Markdown tables for data representation. However, it was excluded because it sacrificed depth for visual clarity. Its analysis of psychological dynamics remained superficial, and it occasionally utilized generic SARA terminology (e.g., "Factor S - Safety") instead of strictly adhering to the official forensic numbering (Factors 1 to 15) provided by the Graph-RAG context [2].

Llama-3.3-70B-Versatile [6] emerged as the absolute winner of the benchmark. It provided the best balance between *psychological empathy* and *deterministic legal precision*, correctly mapping Article 612-bis and Article 572 without errors [9].

Despite the 70B model's superior raw performance, **Llama-3.1-8B-Instruct** [6] was selected for the final pipeline and the full-scale evaluation (300 windows). This strategic transition was driven by three fundamental criteria:

1. **Instruction-following and constraint adherence:** the *Instruct* variant is specifically fine-tuned through Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) to strictly follow system prompts. The 8B model proved more "compliant" with the Knowledge Graph's symbolic constraints than the larger 70B models, which occasionally relied too heavily on their internal parametric memory to override retrieved facts.
2. **Validation of the Graph-RAG approach:** the use of an 8B model allows for the demonstration of the architecture's power. It proves that a domain-specific Knowledge Graph can elevate a smaller, more efficient model to match the expert reasoning of a 70B model (achieving 5.00/5.00 in clinical coherence), confirming that the Knowledge Graph is the primary driver of accuracy.
3. **Efficiency and reproducibility:** the 8B scale ensures the system can run on standard GPU resources (e.g., Tesla T4) via 4-bit quantization, maintaining low latency and high stability even in the chaotic Dual-LLM Roleplay scenarios (Strategy 3).

4.4.1. Guided inference

The selected expert knowledge (Tactic definition + SARA criteria + Legal article) is injected into the system prompt of **Llama-3.1-8B-Instruct**. Applying *Knowledge-Controlled Generation* [9], the model is instructed to act as a **Forensic Psychologist and Legal Expert**, providing a structured analysis that must explicitly reference the retrieved graph nodes. This ensures that the final output is a traceable assessment anchored to the provided ontological context.

5. Evaluation and results

5.1. Quantitative analysis (LLM-as-a-Judge)

The full dataset (**300 windows**) was evaluated using an **LLM-as-a-Judge** protocol. **Llama-3.1-8B-Instruct** graded the explanations based on **Grounding**, **Coherence**, and **Legal Accuracy**, achieving an average score of **3.70/5.00**. The detailed breakdown of these global metrics is provided in Table 1.

Table 1
Global Forensic Evaluation Metrics (LLM-as-a-Judge)

Evaluation Metric	Mean Score (1-5)
Factual Grounding	1.09 / 5.00
Clinical Coherence	5.00 / 5.00
Legal Accuracy	5.00 / 5.00
Total Average Score	**3.70 / 5.00**

The most significant results emerged in **Clinical Coherence** and **Legal Accuracy**, where the model reached the **maximum score of 5.00/5.00**. These values confirm that the neuro-symbolic integration successfully eliminates legal hallucinations and ensures total alignment with SARA forensic protocols. Conversely, the score for **Factual Grounding (1.09/5.00)** reflects an inherent evaluation bias: the Judge penalizes the inclusion of technical terminology (legal codes and clinical factors) not literally present in the source text. In a Graph-RAG context, however, this low value quantitatively proves the model's ability to enrich the analysis by *injecting* deterministic domain knowledge from the Knowledge Graph.

5.2. Confidence scoring and forensic caution

In addition to assessing the traditional risk levels, the system implements an explicit **Confidence Score (0-100%)**. The LLM is instructed to evaluate its own degree of certainty based on the quality and specificity of the retrieved graph context. The system maintained a **Mean Confidence of 82.70%**, demonstrating remarkable stability across different generation scenarios, as detailed in Table 2.

Table 2
Performance Analysis by Dataset Generation Strategy

Generation Strategy	Mean Confidence (%)	Judge Score (1-5)
Strategy 1: Single-shot	83.81%	3.71
Strategy 2: Single-LLM Roleplay	82.00%	3.68
Strategy 3: Dual-LLM Roleplay	80.91%	3.70

The data reveals that Strategy 1 reached a peak confidence of **83.81%** due to the didactic nature of the dialogues, while Strategy 3 (Dual-LLM Roleplay) remained robust at **80.91%**, proving that the Knowledge Graph acts as a reliable anchor even in chaotic and highly manipulative conversational contexts. In instances where the Confidence Score is lower than 50%, the system is hard-coded to automatically issue a **Forensic Caution Warning**: *"Ambiguous signal, requires expert human inspection."* This serves as a vital safeguard against hallucinations in sensitive legal and clinical scenarios, improving the Explainable AI (XAI) footprint of the tool.

5.3. Experimental design and ablation study

To rigorously assess the impact of temporal context and symbolic grounding, a controlled ablation study was conducted. This study systematically compared the **Graph-RAG** implementation against a **Vanilla LLM** baseline (parametric only) evaluating due fundamental variables: Temporal Window Sizes and Generation Strategies.

5.3.1. Evaluation metrics

The system's performance was measured using an **LLM-as-a-Judge** protocol based on four forensic criteria, each scored on a 1-5 scale to ensure a multidimensional validation of the neuro-symbolic pipeline:

1. **Factual grounding:** accuracy in referencing legal (Codice Rosso) and clinical (SARA) nodes retrieved from the Knowledge Graph.
2. **Clinical coherence:** consistency of the identified psychological dynamic with established SARA risk assessment protocols.
3. **Legal precision:** correct application of the penal articles to the detected behaviors.
4. **Sensitivity to escalation (delta analysis):** measured as the inverse of performance variance between narrow (3 turns) and wide (10 turns) windows. A high score indicates maximum robustness against "context noise" and conversational complexity.

The global results for these metrics are summarized in Table 3, highlighting the system's absolute precision in clinical and legal reasoning.

Table 3
System Performance on Primary Forensic Metrics

Metric	Judge Score (1-5)
Factual Grounding	1.09
Clinical Coherence	5.00
Legal Accuracy	5.00
Forensic Stability (Delta Analysis)	4.99
Total Mean Score	**3.70**

5.3.2. Findings

The study demonstrated that as window sizes increased from 3 to 10, the **Vanilla LLM** baseline exhibited significant "context drift," frequently hallucinating penal articles or generating overly generic psychological profiles. Conversely, the **Graph-RAG** approach maintained exceptional stability in reasoning. While the *Factual Grounding* score reflected the strictness of the evaluation protocol regarding external knowledge injection, the *Clinical Coherence* and *Legal Precision* remained at peak levels (**5.00/5.00**) even in long-sequence contexts. This proves that the integration of the Knowledge Graph acts as a robust anchor, ensuring that the model's reasoning is derived from deterministic forensic definitions rather than probabilistic weight patterns.

Table 4
Ablation Results: Strategy Stability and Variance (Delta)

Strategy	Confidence	Total Score	Delta (from S1)
Strategy 1: Single-shot	83.81%	3.71	-
Strategy 2: Single-LLM Roleplay	82.00%	3.68	-0.03
Strategy 3: Dual-LLM Roleplay	80.91%	3.70	-0.01

The Delta Analysis confirms this stability, showing a negligible variance between the didactic Strategy 1 and the chaotic Strategy 3 (Delta = -0.01).

6. Conclusions and future work

We successfully developed a neuro-symbolic pipeline for CIPV detection that is explainable and clinically valid. By anchoring the reasoning of Large Language Models to a deterministic Knowledge Graph, we significantly reduced hallucinations and provided a traceable forensic path for legal and clinical assessment.

Our future work will focus on the following key areas:

- **Human-in-the-loop validation:** collaborating with professional forensic psychologists to perform a "blind review" of the automated reports, ensuring that the AI's clinical reasoning aligns with human expertise.
- **Multimodal integration:** extending the Graph-RAG framework to process multimodal inputs, such as images, emojis, and voice notes, which often carry significant weight in digital coercive control.
- **Longitudinal pattern detection:** moving beyond short-term sliding windows to analyze long-term behavioral changes across months of conversation, allowing the graph to model the progressive "grooming" and "isolation" phases of IPV.
- **Privacy-preserving architectures:** exploring the deployment of Small Language Models (SLMs) on edge devices via Federated Learning. This would allow for local, on-device detection of abuse, ensuring that highly sensitive victim data never leaves the personal device, thus aligning with the highest ethical and safety standards.
- **Dynamic knowledge updates:** integrating automated KG extraction techniques [15] to keep the knowledge base updated with new legislative reforms and evolving digital threats (e.g., deepfake extortion patterns).

A. Ontological schema

This appendix provides the exhaustive list of all nodes and relationships encoded within the Neo4j Knowledge Graph. This codebook represents the deterministic constraints provided to the LLM during the Graph-RAG inference phase.

A.1. Category nodes (psychological archetypes)

These nodes represent the systemic relational dynamics within abusive partnerships:

1. **Narcissist and submissive:** a dynamic where the abuser demands constant admiration and dominance, while the victim exhibits total compliance to avoid conflict.
2. **Controller and isolated:** the abuser systematically cuts off the victim from support networks to foster absolute dependency.
3. **Jealous-Obsessive and Submissive:** characterized by unfounded accusations of infidelity and continuous surveillance, resulting in the victim's withdrawal or submission.
4. **Sadistic-cruel and masochistic:** iIntentional infliction of psychological or physical pain, where the victim is conditioned to endure suffering without rebellion.
5. **Victimizer and rescuer (savior complex):** the abuser feigns vulnerability to exploit the victim's psychological need to "fix" or care for the partner.
6. **Manipulator and emotionally dependent:** emotional blackmail and threats of abandonment targeting a victim terrified of loneliness.
7. **Violent and submissive:** a relationship dominated by vulgar language and physical threats used to enforce strict abuser-imposed rules.
8. **Perfectionist critic and chronically insecure:** persistent denigration focused on maternal or social roles, forcing the victim to perpetually seek validation.

9. **Dominant and emotional slave:** an absolute command structure where the victim constantly apologizes and minimizes their own worth.
10. **Psychopath and flatterer:** calculated manipulation involving Gaslighting and deceit, where the abuser masks lack of empathy with grandiosity.

A.2. Tactic nodes (abuse mechanisms)

Specific coercive behaviors identified within the communication patterns:

1. **Jealousy:** accusations of infidelity and obsessive monitoring of social interactions or mobile devices.
2. **Location control:** GPS tracking, unexpected home/work appearances, and real-time surveillance of movements.
3. **Isolation:** restricting access to friends or family members to weaken the victim's social fabric.
4. **Economic control:** seizure of financial resources, forbidding employment, or controlling personal expenditures.
5. **Threats:** verbalized intentions to cause physical harm, self-harm (suicide blackmail), or harm toward children.
6. **Gaslighting:** distorting reality to make the victim question their memory, sanity, or perception of events.
7. **Humiliation:** private or public ridicule, insulting intelligence or appearance, and systematic belittlement.
8. **Sexual coercion:** non-consensual sexual acts, reproductive coercion, or blackmail via intimate imagery.
9. **Blame Shifting:** projecting the abuser's violent behavior onto the victim ("You forced me to act this way").
10. **AI manipulation (Deepfake):** illicit creation or threat to spread AI-generated synthetic media to extort or degrade the victim.

A.3. Stage nodes (relational lifecycle)

Temporal contexts of abuse progression:

1. **Bonding and integration:** the initial phase; control often surfaces as over-attentiveness or "love-bombing."
2. **Differentiation:** rising tension as the victim attempts to regain autonomy, met by abuser coercion.
3. **Avoidance and breakdown:** communication ceases or becomes purely hostile; fear of abuser loss of control intensifies.
4. **Final phase and peril:** separation or termination phase; the stage with the highest clinical risk of stalking and lethal violence.

A.4. SARA nodes (clinical risk factors)

Coded criteria from the *Spousal Assault Risk Assessment* used for forensic risk determination:

1. **SARA Factor 2:** severe threats of violence or evidence of intent to cause death/serious harm.
2. **SARA Factor 3:** recent escalation in the frequency or severity of threats or physical attacks.
3. **SARA Factor 5:** abuser exhibits negative attitudes (minimization or total denial) toward the abuse.
4. **SARA Factor 7:** significant conflict regarding separation, divorce, or custody disputes.
5. **SARA Factor 12:** extreme terror expressed by the victim, paralyzing self-protection mechanisms.

A.5. Law nodes (penal code articles)

Legal provisions from the Italian "Codice Rosso" regulatory framework:

1. **Art. 572 P.C.:** domestic ill-treatment and habitual psychological/physical abuse within a household.
2. **Art. 612-bis P.C.:** stalking behaviors causing persistent anxiety or alteration of life habits.
3. **Art. 610 P.C.:** private violence; forcing another to tolerate, perform, or omit an act via coercion.
4. **Art. 612-ter P.C.:** non-consensual dissemination of sexually explicit imagery (Revenge Porn).
5. **Art. 612-quater P.C.:** illicit distribution of synthetic media generated via AI (AI-based Harassment).

A.6. Structural relationships

The Graph-RAG system uses the following edge types to derive expert-grounded insights:

- **CHARACTERISTIC_OF / TYPICAL_OF:** maps an abuse *Tactic* to a psychological *Category* (e.g., *Gaslighting* is *CHARACTERISTIC_OF Psychopath and Flatterer*).
- **POTENTIALLY_CONFIGURES:** maps a behavioral *Tactic* to a legal *Law* node (e.g., *Jealousy POTENTIALLY_CONFIGURES Art. 612-bis P.C.*).
- **TRIGGERS_FACTOR:** identifies how specific behaviors activate *SARA Factor* risk indicators (e.g., *Threats TRIGGERS SARA Factor 2*).
- **EVOLVES_IN / CRITICAL_RISK_IN:** represents the trajectory of abuse toward high-risk *Stage* nodes like *Final Phase and Peril.*
- **MANIFESTS_DURING:** associates a *Tactic* with its most likely *Stage* in the relationship lifecycle.
- **CAUSES:** links a behavior or stage directly to the onset of severe outcomes like extreme terror.

B. System prompts

This appendix provides the technical configuration and system prompts utilized by the LLMs during the different phases of the pipeline.

B.1. Entity extraction prompt

The following prompt was used by the lightweight Llama-3.1-8B to extract key entities for the Hybrid Search (Entity Linking) phase:

```
Analyze the following chat segment and extract 3 to 5 key terms
related to abuse tactics (e.g., GPS, password, money, friends,
jealousy, threat, permission).

CHAT: "{chat_context}"

Respond ONLY with a comma-separated list of terms in Italian.
```

B.2. Vanilla LLM baseline prompt

The prompt used for the ablation baseline (Vanilla LLM), which relies exclusively on the model's parametric knowledge without any Knowledge Graph grounding. We utilized the official Llama-3 chat template to separate system instructions from the user task:

```
messages = [
  {"role": "system", "content": "You are an expert Forensic Psychologist and Legal Expert
      specialized in Intimate Partner Violence (IPV) and Italian Law ('Codice Rosso'). You provide
       highly structured, objective assessments."},
  {"role": "user", "content": """
```

```
   EVALUATE THIS CHAT SEGMENT: "{chat_context}"

   Provide a structured forensic report containing EXACTLY these 5 points:
   1. PSYCHOLOGICAL DYNAMIC: Identify specific abusive patterns or archetypes.
   2. LEGAL ANALYSIS: Cite relevant Italian Penal Code articles (e.g., Art. 572, 612-bis).
   3. SARA PROTOCOL ASSESSMENT: Identify applicable clinical risk factors.
   4. GLOBAL RISK LEVEL: Categorize danger as Low, Medium, High, or Critical.
   5. CONFIDENCE SCORE: Provide a numeric percentage (0-100%). If below 50%, you MUST add: "
       WARNING: Ambiguous signal, requires expert human inspection."

   Respond entirely in Italian.
   """}
]
```

B.3. Graph-RAG inference prompt

The prompt used in the Graph-RAG pipeline, where the model's reasoning is strictly constrained by the retrieved structured knowledge. It also employs the Llama-3 chat template for strict task compliance:

```
messages = [
  {"role": "system", "content": "You are an expert Forensic Psychologist. GROUND your response
      STRICTLY on the Graph Knowledge provided."},
  {"role": "user", "content": """
   CHAT LOG: "{chat_context}"

   EXPERT GRAPH KNOWLEDGE:
   {knowledge_summary}

   Analyze dynamic, laws, SARA factors, risk level, and confidence (0-100%). Respond in Italian.
   """}
]
```

B.4. Evaluation protocol (LLM-as-a-Judge)

The protocol used to validate the output quality, ensuring objective assessment of the generated forensic reports:

```
You are an expert Forensic Psychology Professor. Grade the following analysis on a scale of 1-5:
1. GROUNDING: Does it correctly reference the provided graph knowledge?
2. COHERENCE: Is the psychological dynamic consistent with the SARA protocol?
3. LEGAL ACCURACY: Is the application of the Italian Penal Code correct?

CHAT CONTEXT: "{context}"
ANALYSIS TO EVALUATE: "{analysis}"

OUTPUT FORMAT (JSON ONLY, NO MARKDOWN):
{"grounding": 0, "coherence": 0, "legal_accuracy": 0}
```

References

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